LIS 5775: Final Presentation Project

Securing the Cloud

A Phased Defense Against a Rising Tide of Cyber Threats

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Introduction & Project Scope

1. As cloud computing becomes the backbone of enterprise infrastructure, traditional perimeter-based defenses fall short in protecting against modern cyber threats.
2. The rapid shift to the cloud has expanded the attack surface—exposing organizations to unauthorized access, data exfiltration, lateral movement, and persistent threats.
3. In 2023, cloud security breaches rose 27%, with data loss and unauthorized access among the top attack vectors (Süß et al., 2023).

Research Question: How can secure communication, network traffic analysis, threat detection, and incident response techniques be effectively implemented in a cloud environment to enhance security posture and mitigate cyber threats?

Project Scope – implemented and tested four critical phases:

i. Secure Communication
ii. Network Traffic Analysis & Anomaly Detection
iii. Threat Detection & Response
iv. Incident Response & Recovery

Literature Review: Threat Environment & Foundations

- Di Sano et al. (2023): Traditional perimeter-based models are inadequate; Zero Trust andcontinuous monitoring are essential.
- Hashizume et al. (2013): Cloud vulnerabilities require multilayered defense.
- Süß et al. (2023): Integrated tools like Zeek and Snort enhance behavioral analytics.
- Boyle & Panko (2023): Tools like Wireshark provide forensic insight into real-time trafficanomalies.
- Patterson et al. (2023): Sysmon is essential for forensic log analysis.

Literature Review: Phase Alignment & Threat Types

1. Phase 1 (Secure Communication): Lack of encryption is a leading cause of data exfiltration. We addressed this using SSH tunneling.

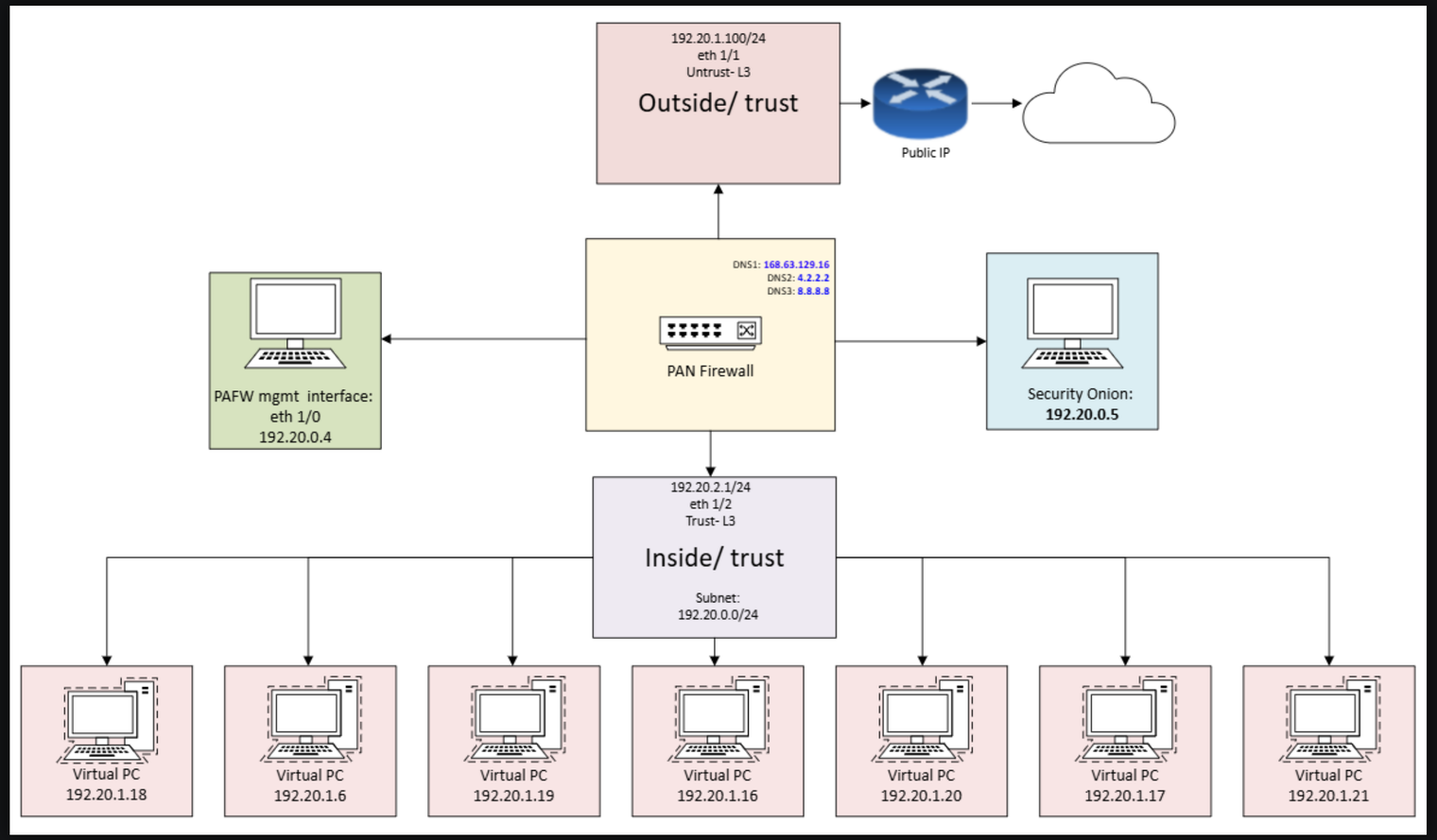
2. Phase 2 (Network Traffic Analysis): SYN flood/DDoS patterns detectable via Zeek logsand Wireshark packet capture.

3. Phase 3 (Threat Detection): Detection of reconnaissance and cleartext vulnerabilitiesusing Wireshark.

4. Phase 4 (Incident Response): Use of Sysmon and Wireshark for post-breachinvestigation.

Computing Environment & Network Topology

- Azure SECNET Labs environment
- VMs: Kali Linux, Windows Server, Security Onion, Snort IDS, OpenSSH Server
- Tools: MobaXterm, Wireshark, Sysmon, OpenSSH, Ansible, Metasploit



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Overview – Phase 1 Secure Communications Practices

- Objective: Establish encrypted communication between subnets using SSH tunneling.
- Threat Environment: Data exfiltration via unencrypted channels.
- Tools: OpenSSH (Server/Client), Wireshark
- Methods: Configured SSH tunnel & validated with file transfers.

Phase 1 – Results & Key Takeaways

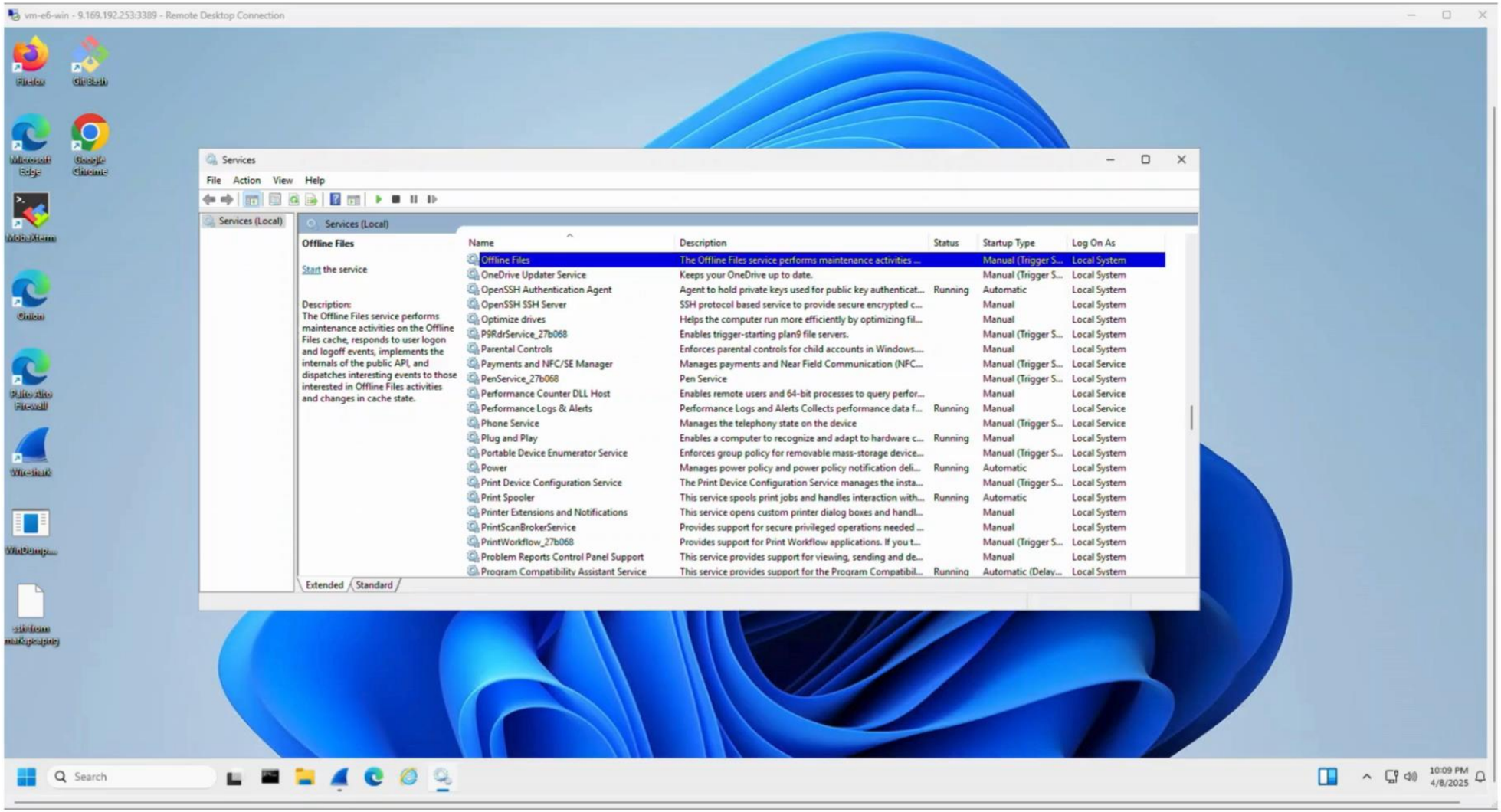
1. SCP used to transfer large payload.

2. Wireshark captured encrypted transmission; clear demonstration of secure channel.

3. Lessons: OpenSSH effective substitute for VPN when firewall/portal access is limited.

4. Future Work: Expand testing with OpenVPN or StrongSwan.

Phase 1 - Secure Communications Demo



Overview – Phase 2 Network Traffic Analysis & Anomaly Detection

- Objective: Detect SYN flood DDoS using Wireshark & Security Onion.
- Threat: High-volume, incomplete SYN packet floods.
- Tools: Kali Linux (hping3), Ansible, Wireshark, Security Onion
- Methods: Automated SYN flood, packet capture, and log analysis.

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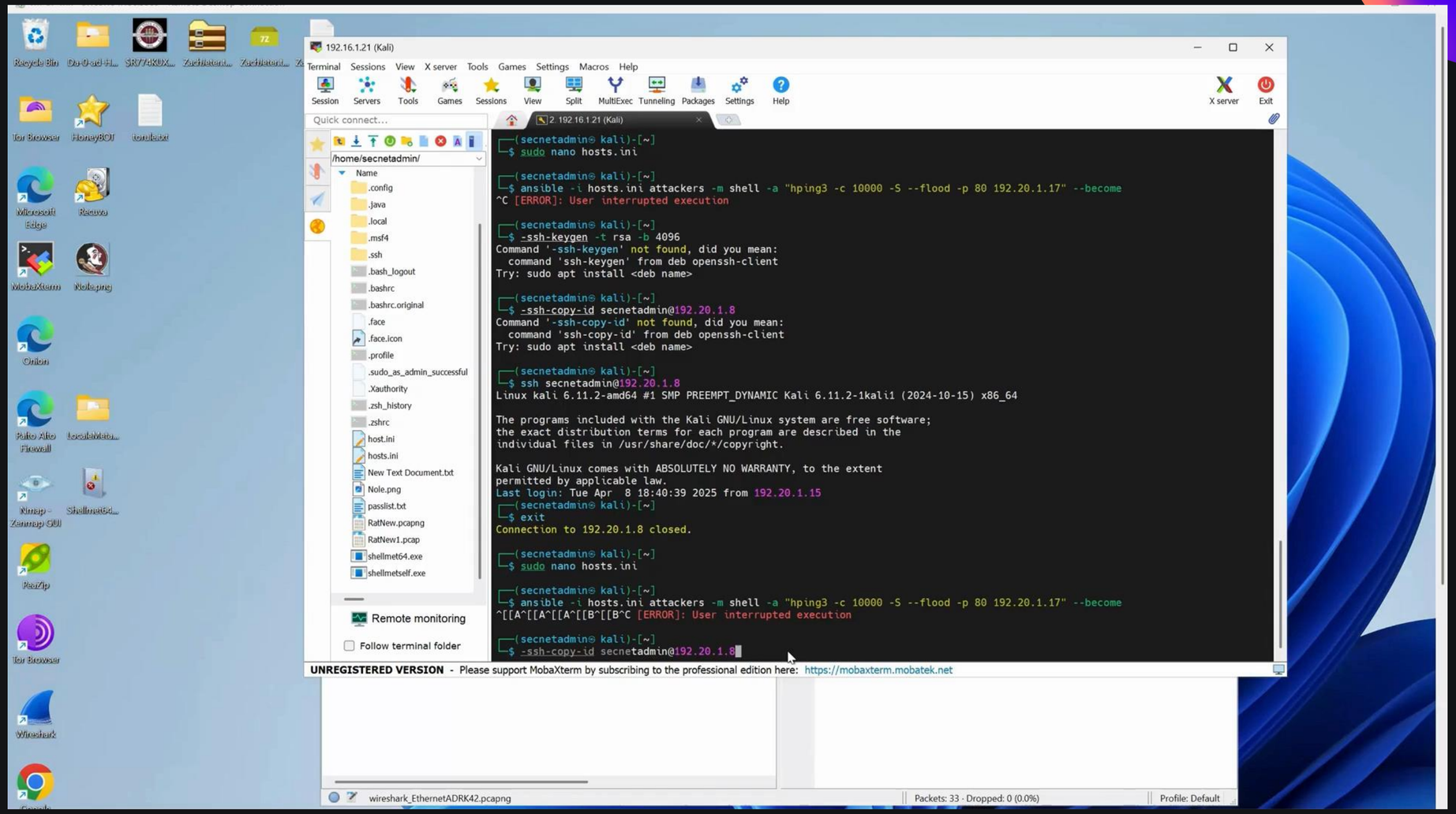
Phase 2 – Results & Key Takeaways

1. Zeek flagged spikes as Zeek.Weird; Wireshark captured SYN flood patterns.

2. Lessons: Visual and log-based anomaly detection are complementary.

3. Future Work: Introduce rate-limiting firewall rules.

Phase 2 – Network Traffic Analysis & Anomaly Detection Demo



- Objective: Detect cleartext credentials and reconnaissance scans.
- Threat: Exploitation of unencrypted web traffic; port scanning.
- Tools: Wireshark
- Methods: Captured HTTP traffic, filtered for GET/POST login packets.

03 Overview – Phase 3 Threat Detection & Response

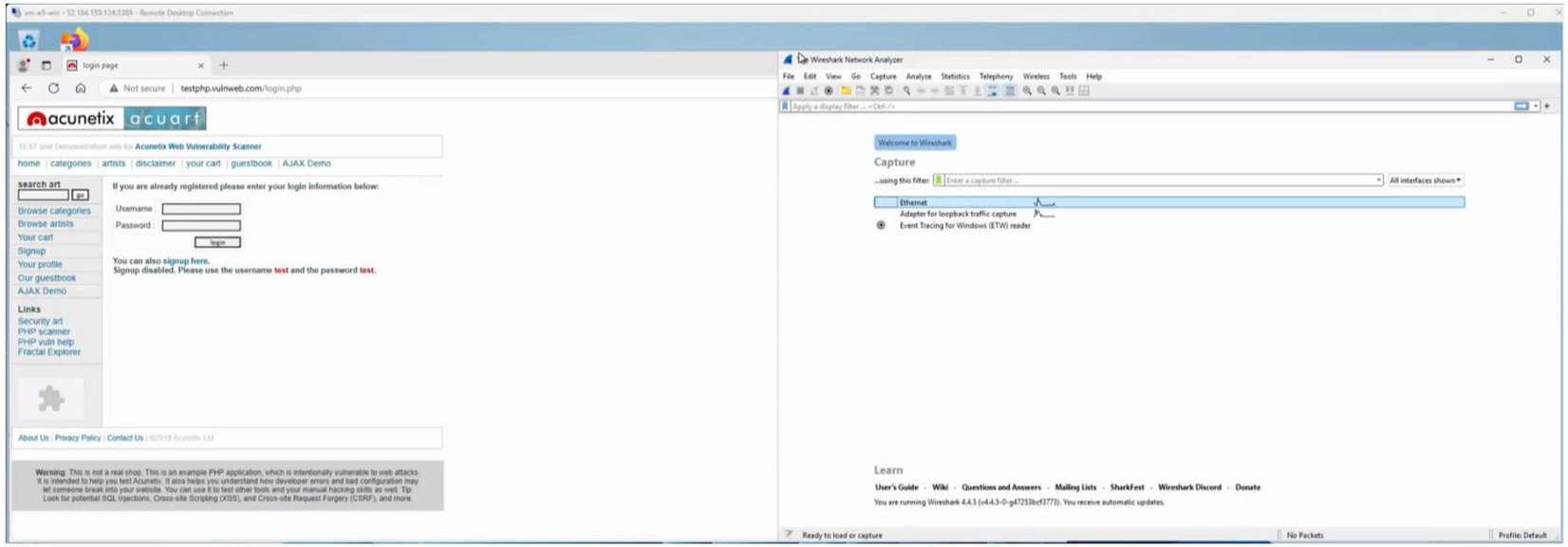
Phase 3 – Results & Key Takeaways

1. Cleartext login credentials captured via Wireshark.

2. Successful secure copy from VM to OpenSSH Server.

3. Lessons: Wireshark is effective for IDS under constrained lab setups.

Phase 3 – Threat Detection & Response Demo



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Overview – Phase 4 Incident Response & Recovery

- Objective: Simulate RAT attack, log and analyze system compromise.
- Threat: Remote Access Trojans allow unauthorized control.
- Tools: Msfvenom, Metasploit, Wireshark, Sysmon
- Methods: Reverse shell payload creation, logging via Sysmon.

Phase 4 – Results & Key Takeaways:

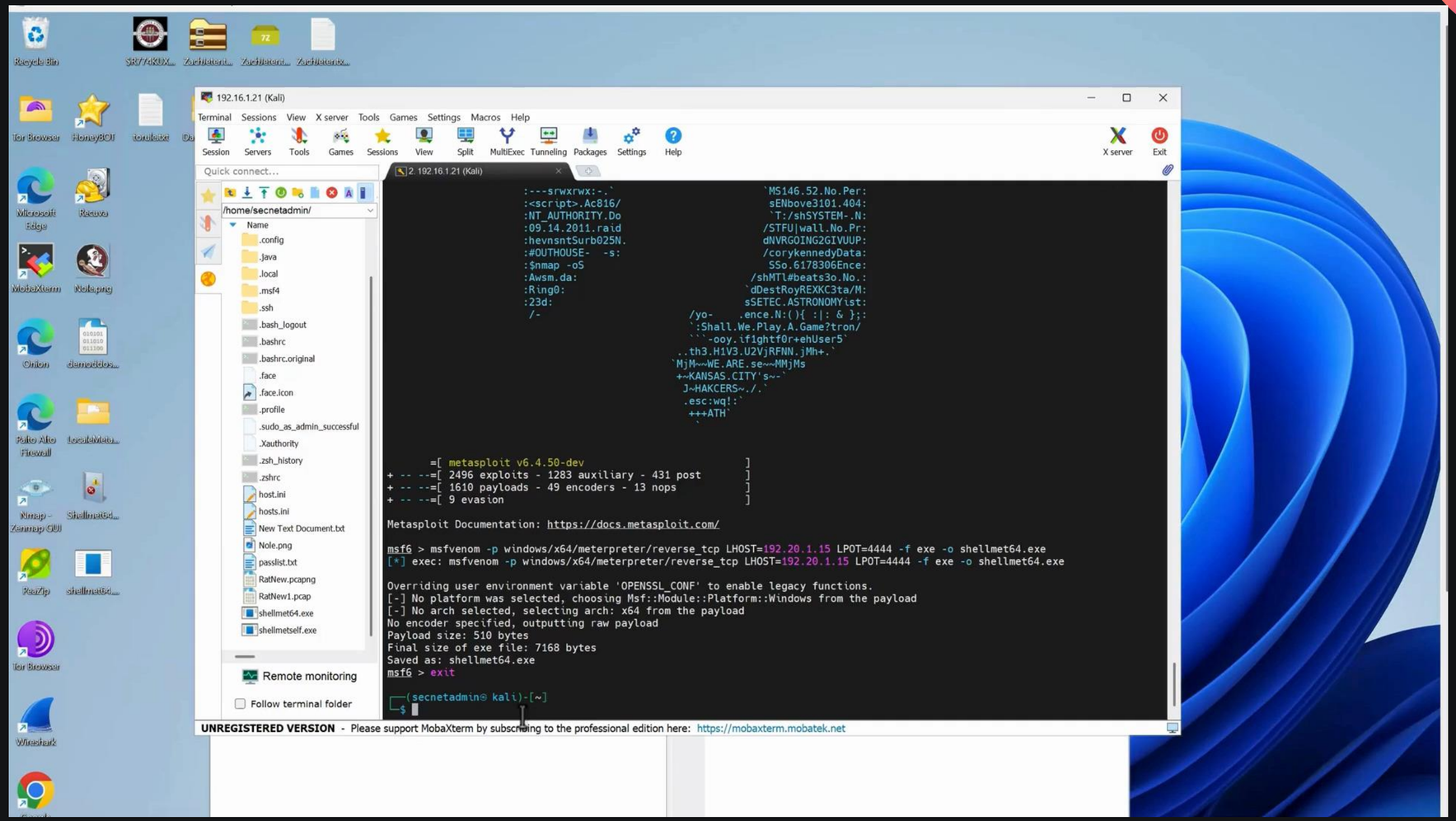
1. Wireshark: Captured outbound RAT traffic.

2. Sysmon: Logged process creation and network connections.

3. Lessons: Host-based and network monitoring are complementary.

4. Future Work: Use of centralized SIEM (e.g., Splunk) for alert correlation.

Phase 4 – Incident Response & Recovery Demo



Conclusion – Final Reflections

Cloud threats require layered defenses: encrypted communication, anomaly detection, intrusion monitoring, and forensic readiness.

All 4 phases successfully executed using open-source tools.

Addressed gaps from midterm: added threat environment rationale, testing validation, structured analysis.

Continuous monitoring, actionable intelligence, and clear response workflows are essential for cloud resilience.

Thank You

Any Questions?